

XINGREN, China

WMO: 579020

Lat: 25.4308N

Lon: 105.1832E

Elev: 4524

StdP: 12.45

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	31.0	32.9	24.5	22.0	45.7	27.4	25.2	42.7	19.7	64.6	16.2	63.8	3.7	40	0.334

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	85.5	66.8	83.6	67.5	81.9	67.6	72.3	79.5	71.6	78.8	70.8	77.7	9.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.3	132.2	75.2	69.5	128.7	74.6	68.9	125.8	74.0	39.5	80.3	38.7	79.2	38.0	77.8	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	13.4	11.3	28.8	90.0	2.0	2.3	27.4	91.6	26.2	92.9	25.0	94.2	23.6	95.8
			27.6	74.1	2.2	0.9	26.0	74.7	24.7	75.2	23.5	75.7	21.8	76.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	61.4	45.0	49.7	56.5	64.6	68.9	71.5	73.2	72.5	68.8	62.7	55.8	47.2
	DBStd	11.71	8.03	10.04	8.65	7.65	5.65	4.15	2.77	3.05	4.48	5.59	6.44	6.21
	HDD50	515	201	126	35	2	0	0	0	0	0	1	19	131
	HDD65	2437	621	433	287	101	28	4	0	1	16	114	280	552
	CDD50	4684	45	118	236	441	586	645	720	699	564	393	194	43
	CDD65	1132	0	5	23	90	149	199	255	234	130	42	5	0
	CDH74	5234	1	53	208	632	805	833	1058	1026	507	96	14	1
	CDH80	863	0	4	35	157	173	123	149	163	57	2	0	0
Wind	WSAvg	4.8	4.8	5.6	5.7	5.8	5.2	4.6	4.6	3.9	4.0	4.1	4.5	4.4
Precipitation	PrecAvg	52.4	0.8	1.0	1.1	2.9	6.8	10.2	8.8	8.4	5.4	4.2	2.0	0.9
	PrecMax	74.4	1.6	3.1	4.1	5.8	12.2	18.7	18.7	16.5	14.3	16.0	5.7	3.3
	PrecMin	28.8	0.4	0.1	0.2	0.1	1.7	3.2	2.2	2.5	1.4	1.0	0.4	0.1
	PrecStd	8.9	0.3	0.7	0.8	1.5	2.6	4.1	4.3	3.9	2.8	3.0	1.3	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.4	79.9	84.2	88.1	88.7	86.0	85.6	86.1	84.7	79.6	75.9	69.9
		MCWB	51.7	52.8	56.8	60.0	63.8	70.8	70.9	69.9	69.2	66.0	60.1	54.4
	2%	DB	66.0	75.0	79.7	84.4	84.3	83.3	83.5	83.6	81.5	76.6	72.0	64.5
		MCWB	52.1	53.1	56.5	60.7	65.9	70.2	70.5	69.9	68.0	64.7	59.5	53.8
	5%	DB	61.5	70.7	75.5	81.1	81.4	81.1	81.7	81.8	79.3	74.3	68.8	60.3
		MCWB	51.5	53.2	57.2	61.6	65.9	69.9	70.2	69.6	67.3	63.8	58.5	52.2
	10%	DB	56.7	65.9	70.7	77.6	78.5	78.9	79.7	79.8	77.0	71.4	65.8	56.5
		MCWB	50.3	53.0	57.1	61.9	65.9	69.3	69.8	69.3	66.6	63.2	57.7	50.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.6	59.1	62.6	67.9	71.1	73.3	73.4	72.8	71.3	68.2	64.1	57.1
		MCDB	64.3	70.6	72.5	77.4	79.5	80.5	80.1	80.7	78.7	75.3	70.6	64.6
	2%	WB	53.6	56.9	60.7	66.3	69.9	72.2	72.2	71.8	70.1	66.8	61.9	54.8
		MCDB	61.9	66.9	70.8	75.6	78.3	79.4	79.0	79.5	77.1	73.1	67.8	61.7
	5%	WB	52.1	55.2	59.4	64.9	68.9	71.3	71.5	70.9	69.2	65.7	60.4	53.1
		MCDB	59.3	64.6	69.3	74.1	76.7	78.3	78.3	78.3	75.7	71.4	66.1	59.6
	10%	WB	50.4	53.4	57.9	63.6	67.8	70.4	70.7	70.1	68.3	64.5	58.7	51.0
		MCDB	56.6	62.8	67.5	72.5	75.0	76.7	77.2	77.0	74.3	69.9	64.4	55.9
Mean Daily Temperature Range	5% DB	MDBR	11.1	13.6	15.0	15.2	13.7	11.2	11.4	12.7	12.5	10.7	12.2	10.8
		MCDBR	21.3	25.1	24.5	21.7	18.7	14.8	14.9	16.6	16.3	15.2	17.6	18.8
		MCWBR	9.7	9.0	8.2	6.8	6.3	5.2	4.9	5.3	5.2	5.5	7.1	9.2
	5% WB	MCDBR	18.4	21.0	20.4	18.5	14.5	12.4	12.4	13.7	12.9	12.5	13.9	16.7
MCWBR		9.2	9.3	8.7	7.1	5.9	5.2	4.8	5.2	4.9	5.2	6.1	8.6	
Clear-Sky Solar Irradiance	taub	0.491	0.488	0.620	0.588	0.500	0.495	0.462	0.504	0.508	0.551	0.519	0.528	
	taud	1.855	1.898	1.624	1.746	2.057	2.110	2.207	2.062	2.013	1.824	1.823	1.761	
	Ebn at Noon	234	249	223	235	256	256	264	253	249	227	224	215	
	Edh at Noon	57	59	82	74	54	51	46	53	55	62	59	61	
All-Sky Solar Radiation	RadAvg	724	1048	1245	1498	1510	1315	1497	1448	1234	895	917	672	
	RadStd	162	222	138	181	155	172	107	111	131	127	117	114	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.68	N/A	N/A	+1.03	N/A	N/A	N/A	N/A	+212	+122
(47)	Regional (5 neighbors)	+0.77	N/A	N/A	+1.08	+0.36	N/A	N/A	-198	+229	+90

Nomenclature: See separate page